

Database Systems

LAB#10



23K-2001

BCS-5J

Q1:

```
CREATE TABLE product_inventory (  
    product_id NUMBER PRIMARY KEY,  
    product_name VARCHAR2(50),  
    stock NUMBER,  
    price NUMBER  
);  
  
INSERT INTO product_inventory VALUES (1, 'Keyboard', 50, 1500);  
INSERT INTO product_inventory VALUES (2, 'Mouse', 30, 800);  
INSERT INTO product_inventory VALUES (3, 'Headset', 20, 2500);  
  
UPDATE product_inventory  
SET stock = stock - 5  
WHERE product_id = 1;  
  
SAVEPOINT stock_update;
```

Q2:

```
INSERT INTO employees (employee_id, first_name, last_name, salary)  
VALUES (3001, 'Ali', 'Khan', 50000);  
  
UPDATE employees  
SET salary = salary * 1.10  
WHERE employee_id = 3001;  
  
SAVEPOINT salary_increase;  
  
UPDATE employees  
SET salary = salary * 1.05  
WHERE employee_id = 3001;  
  
ROLLBACK TO salary_increase;
```

Q3:

```
BEGIN
  INSERT INTO customer (customer_id, customer_name)
  VALUES (501, 'Hamza');

  INSERT INTO orders (order_id, customer_id, amount)
  VALUES (9001, 501, 4500);

  COMMIT;
EXCEPTION
  WHEN OTHERS THEN
    ROLLBACK;
END;
```

Q4:

```
SET AUTOCOMMIT ON;

INSERT INTO sales (sales_id, customer_id, amount)
VALUES (101, 501, 3000);
SELECT * FROM sales WHERE sales_id = 101;

SET AUTOCOMMIT OFF;
```

Q5:

```
UPDATE transactions SET balance = balance - 500 WHERE account_id = 10;  
SAVEPOINT sp1;
```

```
UPDATE transactions SET balance = balance + 200 WHERE account_id = 10;  
SAVEPOINT sp2;
```

```
UPDATE transactions SET balance = balance - 300 WHERE account_id = 10;  
SAVEPOINT sp3;
```

```
ROLLBACK TO sp2;
```